

Elliot Bentham

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SUMMARY

Quantitative Developer with expertise in Python and a strong understanding of UK and European Power and Gas markets, experienced in power stack and dispatch modelling, demand forecasting and balancing mechanics. Building a personal medallion-architecture research platform for European power markets, hosted on GitHub and integrating data from Elexon, ENTSO-E, ENTSO-G and weather sources, with a React + Typescript front-end over the analytics layer. Developed to support my own research in energy markets, including power stack modelling, as well as to develop fluency in agentic AI development workflows. Adept with modern data engineering practices such as cloud engineering (AWS, Azure), CI/CD, distributed computing and end-to-end ML workflows. Currently at ICBCS working across FICC asset classes (commodities, rates, credit, FX, volatility) developing tools, data workflows and risk analytics for trading and risk management.

WORK EXPERIENCE

Quantitative Developer, ICBCS

September 2023 – Present

- Designed and implemented an end-to-end application replicating the vendor risk management system to enhance transparency, debugging capability, and analytical granularity. The solution comprised a Python ETL and analytics engine, integrating Value-at-Risk calculations, stress testing, and options pricing, leveraging high-performance parallel computing for efficient batch processing. Results are delivered to traders and risk managers through an interactive Power BI front end, contributing towards the bank's wider move away from Excel-based processes towards more reliable and auditable tooling.
- Played a pivotal role in transitioning from a legacy data storage system to a new infrastructure, facilitating the migration of time series across multiple asset classes, including IR curves, CDS, FX, bonds, commodities, and vol surfaces. Responsible for time series proxying, curve stripping (IR and CDS), and applying data smoothing techniques for proxied time series and volatility surfaces.
- Undertaking Multiverse Advanced Data Fellowship (BSc Hons equivalent). Specialising in cloud-based data pipeline architecture, scalable machine learning workflows, data governance, and modern cloud platforms (AWS, Azure), with applied projects in end-to-end ML model development and deployment.

Junior Quantitative Analyst, RISQ

December 2022 – March 2023

- Designed and implemented a Python ETL pipeline to ingest, validate, and analyse energy market data from ENTSO-E, ENTSO-G, Elexon and major weather providers (Met Office, ECMWF), alongside other open-source APIs. Architected for analytical readiness and downstream model consumption.
- Developed a suite of demand forecasting models for European power and gas, including GAM-based approaches, for use in power stack modelling and short-term trading analysis.
- Built strong fundamentals in European power and gas markets — key drivers, traded products, exchange mechanisms, and the interaction between physical and financial market dynamics.

Software Engineer, Bank of America Merrill Lynch

January 2022 – December 2022

- Worked within the FX Options desk team, building and maintaining Python-based pricing and risk models for FX option products in BAML's proprietary Quartz platform.
- Implemented derivatives pricing methodologies in production code — including Black-Scholes analytical pricing and Monte Carlo simulation — applied to FX option valuation and risk reporting.
- Worked with SQL and NoSQL data stores via the Quartz framework, integrating market data, trade capture, and model output into trader-facing analytics.

EDUCATION

M.Sc. Computational Finance, University College London, Merit

2020-2021

Relevant Modules include:

- Algorithmic Trading
- Financial Market Modelling and Analysis
- Market Risk and Portfolio Theory
- Machine Learning with Applications in Finance

Project title: Analysis of Feature Selection Techniques for Trading Signal Generation

Undertook project in collaboration with Banking Science Ltd. Used distributed computing (PySpark and SQL) in the Jupyter Notebook environment to employ a range of feature selection techniques and predictive models to identify informative factors and generate trading signals. Final signals back tested using a mock trading strategy to replicate live signal performance, using expected shortfall, Sharpe ratio and other metrics to assess the performance of the signals.

B.Eng. Electrical and Electronic Engineering, Newcastle University, First Class Honours

2017-2020

Relevant Modules include:

- Digital Signal Processing
- Linear Controller Design and State Space Analysis

Project title: A New Approach to Data Encryption using Number Theoretic Transforms

Coded a novel encryption algorithm in C++ using the Mersenne Number Transform, utilizing the avalanche effect of Number Theoretic Transforms to increase the security of encryption schemes. Received an excellent grade for the presentation of my algorithm and corresponding research to the supervising professor and fellow.

A-Levels, Durham Johnston Sixth Form

2015-2017

A-Levels in Mathematics, Further Mathematics and Physics. AS-Level in Computer Science.

EXPERIENCE AND INTERESTS

- **Gridflow (open-source, github.com/EBentham/gridflow):** Building an open-source Python data pipeline and quant model library for UK and European power and gas markets. Architected around a medallion design to ingest, validate and analyse data from Elexon BMRS, ENTSO-E, ENTSO-G, GIE and NESO carbon intensity, with point-in-time correctness enforced at the SQL layer for backtesting integrity. Developing a suite of forecasting and optimization models, including LightGBM quantile regression with conformal prediction for probabilistic demand forecasting, and linear programming approaches for power stack dispatch. Development workflow leans deliberately on agentic AI tooling - applying context engineering, spec-first prompting and verification-gated code generation throughout.
- **Energy markets and decarbonization:** Particularly interested in how energy systems will develop over the next few years, as electrification accelerates through AI data centers, EVs and electric heat pumps alongside increasing renewable penetration. This shift will reshape energy markets, with prices becoming increasingly weather driven and value moving towards flexibility, storage and demand response.
- **Interest in FinTech:** I am a member of the FinTech society at UCL, Europe's largest financial technology society. I competed in a competition organized by the society (in partnership with Goldman Sachs) to design an automated trading algorithm on the Quant Connect platform, coded in Python. I am also a member of the up-and-coming algorithmic trading society.
- **Hobbies:** In my spare time I enjoy working out at the gym and playing video games. I also enjoy hiking and have walked the length of Hadrian's wall.